Radiation fields via Green's function.

Assume monochromatic field $\vec{A}(\vec{r},t) = \vec{A}(\vec{r})\, e^{-i\omega t}$.

Then $\vec{A}(\vec{r})$ satisfies $(k^2 + \vec{\nabla}_r^2)\, \vec{A}(\vec{r}) = -\mu_0 \vec{j}(\vec{r})$

Retarded Green's function satisfies

$$(k^2 + \vec{\nabla}_r^2)\, G(\vec{r}-\vec{r}\,', \omega) = -\mu_0\, \delta(\vec{r}-\vec{r}\,').$$

Therefore

$$\vec{A}(\vec{r}) = \int d\vec{r}\,'\, G(\vec{r}-\vec{r}\,', \omega)\, \vec{J}(\vec{r}\,')$$

Since $(k^2 + \vec{\nabla}_r^2)\, \vec{A}(r) = \int d\vec{r}\,' \underbrace{(k^2 + \nabla_r^2)\, G(\vec{r}-\vec{r}\,', \omega)}_{-\mu_0\, \delta(\vec{r}-\vec{r}\,')}\, \vec{J}(\vec{r}\,')$

$$= -\mu_0\, \vec{J}(\vec{r}) \quad \underline{\text{ok}}.$$

We found previously (lect. 2)

$$G_R(\vec{r},\omega) = \frac{\mu_0}{4\pi r}\, e^{i\frac{\omega}{c} r} = \int_{-\infty}^{\infty} dt\, e^{i\omega t}\, \underbrace{G_R(\vec{r},t)}_{\frac{\mu_0}{4\pi r}\, \delta(t - \frac{r}{c})}$$

Therefore

$$G_R(\vec{r}-\vec{r}\,', \omega) = \frac{\mu_0}{4\pi\, |\vec{r}-\vec{r}\,'|}\, e^{ik|\vec{r}-\vec{r}\,'|}$$

$$\Rightarrow \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int d\vec{r}\,' \, \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|}\, \vec{J}(\vec{r}\,')$$

Now, $|\vec{r}-\vec{r}'| = \sqrt{(\vec{r}-\vec{r}')\cdot(\vec{r}-\vec{r}')} = \sqrt{r^2+r'^2-2\vec{r}\cdot\vec{r}'} =$

$= (\text{assuming } r' \ll r) = r\sqrt{1-\frac{2\vec{r}\cdot\vec{r}'}{r^2}+\left(\frac{r'}{r}\right)^2}$

$$= r\left(1-\frac{\vec{r}\cdot\vec{r}'}{r^2}+\frac{1}{2}\frac{(r')^2}{r^2}-\frac{1}{8}\left(\frac{2\vec{r}\cdot\vec{r}'}{r^2}\right)^2+\dots\right)$$

$$= r-\frac{\vec{r}\cdot\vec{r}'}{r}+\underbrace{\frac{(r')^2}{2r}-\frac{1}{2r}\left(\frac{\vec{r}\cdot\vec{r}'}{r}\right)^2}+\dots$$

Quadratic terms of the expansion can be neglected provided that the phase change in $e^{ik|\vec{r}-\vec{r}'|}$ due to them is much smaller than 2π :

$$\frac{k(r')^2}{r} \ll 2\pi$$

$$r \gg \frac{k}{2\pi}(r')^2 = \frac{(r')^2}{\lambda}$$

where $\lambda = \frac{2\pi}{k}$ is the wavelength of the radiation.

Assuming size of integration over $\vec{r}'$ is l, quadratic terms can be dropped if

$$\frac{l^2}{\lambda} \ll r \quad \text{(Fraunhofer limit)}$$

$$\Rightarrow |\vec{r}-\vec{r}'| \simeq r-\frac{\vec{r}\cdot\vec{r}'}{r} = r-\hat{n}\cdot\vec{r}'$$

where $\hat{n} = \frac{\vec{r}}{r}$ = unit vector along $\vec{r}$.

Additionally, provided $r \gg l$, we can also approximate $\frac{1}{|\vec{r}-\vec{r}'|} \approx \frac{1}{r}$.

Then

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int dr' \, \frac{e^{ikr - ik\hat{n}\cdot\vec{r}'}}{r} \vec{J}(\vec{r}') =$$

$$= \frac{\mu_0}{4\pi} \frac{e^{ikr}}{r} \int dr' \, \vec{J}(\vec{r}') e^{-ik\hat{n}\cdot\vec{r}'} .$$

$$\vec{B}(\vec{r}) = \vec{\nabla} \times \vec{A}(\vec{r}) = \hat{e}_a \varepsilon_{abc} \frac{\partial}{\partial r_b} A_c(\vec{r}) =$$

$$= \frac{\mu_0}{4\pi} \hat{e}_a \varepsilon_{abc} \frac{\partial}{\partial r_b} \left(\frac{e^{ikr}}{r} \int dr' \, J_c(\vec{r}') e^{-ik\hat{n}\cdot\vec{r}'} \right)$$

Derivative can act on e^{ikr}, $\frac{1}{r}$, and $e^{-ik\hat{n}\cdot\vec{r}'}$.

Provided that $\underline{kr \gg 1}$ the main contribution comes from $\frac{\partial}{\partial r_b} e^{ikr} = ik \frac{\partial r}{\partial r_b} e^{ikr}$

But $\frac{\partial r}{\partial r_b} = \frac{\partial \sqrt{\sum_a r_a^2}}{\partial r_b} = \frac{1}{2\sqrt{\sum_a r_a^2}} \underbrace{\frac{\partial}{\partial r_b}\left(\sum_a r_a^2\right)}_{2 r_b} = \frac{r_b}{r} = n_b$

$$\Rightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \hat{e}_a \varepsilon_{abc} n_b (ik) \frac{e^{ikr}}{r} \int dr' \, J_c(\vec{r}') e^{-ik\hat{n}\cdot\vec{r}'}$$

$$= \cancel{} \, ik \, \hat{n} \times \vec{A}(\vec{r})$$

$$\Rightarrow \vec{B}(\vec{r}) \approx ik \frac{\mu_0}{4\pi} \frac{e^{ikr}}{r} \int \hat{n} \times \vec{J}(\vec{r}\,') e^{-ik\hat{n}\cdot\vec{r}\,'} (d\vec{r}\,').$$

This expression applies if

$$kr \gg 1, \quad r \gg \ell, \quad r \gg \frac{\ell^2}{\lambda}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$2\pi \frac{r}{\lambda} \gg 1 \qquad\qquad r\lambda \gg \ell^2$$

$$\Downarrow \qquad\qquad \text{but } r\lambda \ll r^2 \text{ (since } r \gg \lambda)$$

$$r \gg \lambda$$

$$\Downarrow$$

$$r^2 \gg r\lambda \gg \ell^2$$

OR

$$\boxed{r \gg \sqrt{r\lambda} \gg \ell}$$ Far-field approximation.

Once $\vec{B}(\vec{r},\omega)$ is known, we can find

$$\vec{E}(\vec{r},\omega) \text{ from } \underbrace{\vec{\nabla} \times \vec{E}(\vec{r},\omega)}_{\text{as above}} = i\omega \vec{B}(\vec{r},\omega)$$

$$\vec{\nabla} \times \vec{E} = ik\, \hat{n} \times \vec{E}$$

$$\Rightarrow \quad \vec{B} = \frac{k}{\omega} \hat{n} \times \vec{E} \Rightarrow c\vec{B} = \hat{n} \times \vec{E}$$

$$c\vec{B} \times \hat{n} = \hat{n} \times \vec{E} \times \hat{n} = \vec{E}(\hat{n}\cdot\hat{n}) - \hat{n}\overset{0}{\overset{\parallel}{(\hat{n}\cdot\vec{E})}} = \vec{E}.$$